Light

GCSE Computer Science - Ethical, Legal and Environmental Impacts (Set 2)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

00:00 Start Pause Reset

← Back to Legal Ethical Environmental

Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What does 'copyright' protect in the context of digital technology?

- ○ **A.** The right to copy any file you want
- ○ **B.** The legal right of creators to control how their original work is used, copied, and distributed
- ○ **C.** The right to make backup copies of your own files
- ○ **D.** Permission to download free software

Q2 (1 mark)

What is 'open source' software?

- ○ **A.** Software that is always free of charge
- ○ **B.** Software that opens automatically when you start your computer
- ○ **C.** Software whose source code is publicly available and can be freely used, modified, and shared under specific licence terms
- ○ **D.** Software that only works with an internet connection

Q3 (1 mark)

What does the UK GDPR (General Data Protection Regulation) primarily protect?

- ○ **A.** Individuals' rights over how organisations collect, store, and use their personal data
- ○ **B.** Businesses' rights to collect any data they need
- ○ **C.** The government's right to access all personal information
- ○ **D.** Companies' rights to sell customer data freely

Q4 (1 mark)

What is facial recognition technology?

- ○ **A.** Technology that identifies or verifies a person by analysing the unique features of their face
- ○ **B.** Software that edits photographs to make faces look better
- ○ **C.** A camera that only takes photos of faces

- ○ **D.** An app that adds filters to selfies

Q5 (1 mark)

Why might leaving Bluetooth enabled on a mobile phone pose a security risk?

- ○ **A.** Bluetooth uses too much battery power
- ○ **B.** Bluetooth makes the phone too hot
- ○ **C.** Bluetooth interferes with WiFi signals
- ○ **D.** Attackers within range could potentially exploit vulnerabilities to access the device or intercept data

Q6 (1 mark)

What is a key advantage of backing up data to cloud storage?

- ○ **A.** Cloud backups are always faster than local backups
- ○ **B.** Cloud storage is always free
- ○ **C.** Cloud backups are stored in a separate physical location, protecting against local disasters like fire or theft
- ○ **D.** Cloud backups never need updating

Q7 (1 mark)

What is a 'white hat' hacker?

- ○ **A.** A criminal who wears white clothing while hacking
- ○ **B.** Someone who hacks slowly and carefully
- ○ **C.** A security professional who uses hacking skills ethically to find and fix vulnerabilities with permission
- ○ **D.** A hacker who only targets white-coloured websites

Q8 (1 mark)

What privacy concern is specific to smartwatches that monitor health data?

- ○ **A.** The watch might tell the wrong time
- ○ **B.** Sensitive health information like heart rate, sleep patterns, and activity levels is collected and may be shared with third parties
- ○ **C.** The watch screen might be too small to read
- ○ **D.** Smartwatches are too expensive

Q9 (1 mark)

What is a cochlear implant?

- ○ **A.** An electronic device surgically implanted in the inner ear that provides a sense of sound to people with severe hearing loss
- ○ **B.** A device implanted in the eye to improve vision
- ○ **C.** A type of hearing aid worn outside the ear

- ○ **D.** A pacemaker for regulating heartbeat

Q10 (1 mark)

What is a 'delivery drone'?

- ○ **A.** A person who delivers packages very quickly
- ○ **B.** A robot that works inside a warehouse
- ○ **C.** A type of van used for deliveries
- ○ **D.** A remote-controlled or autonomous unmanned aircraft designed to transport packages to customers

Q11 (1 mark)

How do digital activities like streaming video contribute to carbon emissions?

- ○ **A.** They do not - digital activities have no environmental impact
- ○ **B.** Only downloading content uses electricity, not streaming
- ○ **C.** The data centres, network infrastructure, and devices required all consume electricity, much of which is generated from fossil fuels
- ○ **D.** Digital activities only cause pollution if you print something

Q12 (1 mark)

Why might excessive 'screen time' be considered a wellbeing concern for young people?

- ○ **A.** Research links excessive screen use to issues including disrupted sleep, reduced physical activity, and potential impacts on mental health and social development
- ○ **B.** Screens make rooms too bright
- ○ **C.** Screens are too expensive for young people
- ○ **D.** Young people do not enjoy using screens

Q13 (2 marks)

A school installs CCTV cameras throughout its buildings, including corridors, classrooms, and changing areas. What ethical issues does this raise?

- ○ **A.** CCTV cameras are too expensive for schools to afford
- ○ **B.** There are no ethical issues with school surveillance
- ○ **C.** While CCTV can improve safety and deter misbehaviour, cameras in sensitive areas invade student privacy, create an atmosphere of constant surveillance, and raise questions about proportionality - especially for minors who cannot meaningfully consent
- ○ **D.** CCTV only works during daylight hours

Q14 (2 marks)

Food delivery apps track the real-time location of delivery riders. What are the benefits and privacy implications of this tracking?

- ○ **A.** There are only benefits with no privacy implications
- ○ **B.** Delivery apps do not actually use location tracking

○ **C.** Location tracking only benefits the delivery riders themselves

○ **D.** Benefits include customers seeing delivery progress and companies optimising routes; privacy implications include continuous surveillance of workers, detailed records of movements and behaviour patterns, and potential misuse of location data for performance pressure

Q15 (2 marks)

What is 'vendor lock-in' in cloud computing and why might it be a concern for businesses?

○ **A.** When cloud servers are physically locked in secure buildings

○ **B.** When a business becomes so dependent on one cloud provider's specific services and formats that switching to another provider becomes extremely difficult and expensive - creating dependency and limiting future options

○ **C.** When vendors are locked out of their own systems

○ **D.** A security feature that prevents unauthorised access

Q16 (2 marks)

A hospital's computer systems are infected with ransomware that encrypts patient records and demands payment. What makes this attack particularly serious?

○ **A.** Hospitals have the most computers of any organisation

○ **B.** Hospital computers are older than other organisations' computers

○ **C.** The attack affects both patient safety (if staff cannot access medical records for treatment decisions) and patient privacy (sensitive

health data may be stolen and leaked) - lives could be at risk if critical systems are unavailable

○ **D.** Ransomware only affects hospital computers

Q17 (2 marks)

Smartwatches can detect irregular heart rhythms and alert users to potential health problems. What concerns exist about this capability?

○ **A.** Consumer devices may produce false positives causing unnecessary anxiety or false negatives missing real problems; users may rely on unvalidated devices instead of proper medical care; and unclear regulatory status raises questions about medical device standards

○ **B.** Smartwatches are too fashionable for medical use

○ **C.** Heart monitoring uses too much battery power

○ **D.** Only doctors should be allowed to wear smartwatches

Q18 (2 marks)

Researchers are developing brain-computer interfaces that can read neural signals. What unique privacy concerns do these 'neural implants' raise?

○ **A.** Neural implants are too heavy for the brain

○ **B.** The brain does not produce any signals that can be read

○ **C.** Neural implants only work in hospitals

○ **D.** Unlike other data, neural signals could potentially reveal thoughts, emotions, intentions, and memories - raising questions about

'mental privacy' and whether the most intimate aspects of human consciousness should be accessible to technology

Q19 (2 marks)

Companies are testing autonomous lorries that can drive long distances without human drivers. What social and economic impacts might widespread adoption have?

- ○ **A.** Lorries will become smaller
- ○ **B.** Benefits include reduced driver fatigue accidents and lower delivery costs; concerns include potential job losses for millions of professional drivers, economic disruption to communities dependent on trucking jobs, and questions about transition support for displaced workers
- ○ **C.** Autonomous lorries will only operate at night
- ○ **D.** There will be no significant social or economic impacts

Q20 (2 marks)

Research suggests using screens before bedtime can disrupt sleep. What mechanisms are believed to cause this effect?

- ○ **A.** Screens are too heavy to use in bed
- ○ **B.** This effect only occurs with television screens
- ○ **C.** Screens make beds uncomfortable
- ○ **D.** Blue light from screens can suppress melatonin production (the hormone triggering sleep), stimulating content keeps the brain alert, and the habit of checking devices creates anxiety about missing updates - all interfering with the body's natural sleep preparation

Q21 (2 marks)

When a person dies, their family may want access to their digital accounts and files. What legal and practical complications does 'digital inheritance' create?

- ○ **A.** Terms of service may prohibit transferring account access; passwords die with the user; platforms have inconsistent policies about providing access to estates; and private digital content (messages, photos) may contain things the deceased intended to remain private
- ○ **B.** Digital accounts automatically transfer to family members
- ○ **C.** Digital files cannot be deleted after death
- ○ **D.** There are no complications with digital inheritance

Q22 (2 marks)

Some companies use AI systems to screen job applications. What ethical concerns exist about algorithmic hiring?

- ○ **A.** AI systems are too slow for hiring decisions
- ○ **B.** Algorithms always make perfect hiring decisions
- ○ **C.** Algorithms trained on historical hiring data may perpetuate past discrimination; systems may use proxies that correlate with protected characteristics; and the opacity of AI decisions makes it difficult for rejected candidates to understand or challenge why they were excluded
- ○ **D.** Only large companies use algorithmic hiring

Q23 (2 marks)

Virtual Private Networks (VPNs) encrypt internet traffic and can hide users' locations. Why might VPN use raise both legitimate privacy

protection and potential abuse concerns?

- ○ **A.** VPNs make internet connections faster
- ○ **B.** VPNs legitimately protect privacy on public networks, secure business communications, and enable access to information in censored countries - but the same anonymity can enable accessing illegal content, evading geo-restrictions, or hiding criminal activity from investigation
- ○ **C.** VPNs are only used by criminals
- ○ **D.** VPNs do not actually provide any privacy protection

Q24 (3 marks)

A school considers implementing a 'Bring Your Own Device' (BYOD) policy allowing students to use personal smartphones and tablets for learning. Analyse the benefits and challenges this policy would create.

- ○ **A.** Benefits include cost savings (school does not buy devices), students using familiar devices, and preparing students for workplaces where BYOD is common; challenges include digital divide issues (not all families can afford suitable devices), network security risks from unmanaged devices, distraction potential, and technical support complexity for diverse device types
- ○ **B.** BYOD policies have no significant advantages or disadvantages
- ○ **C.** BYOD only affects students who already own devices
- ○ **D.** Schools should never allow personal devices under any circumstances

Q25 (3 marks)

A hospital wants to use an AI system that can analyse medical scans (such as X-rays and MRIs) to help doctors diagnose diseases. Analyse the ethical implications of using AI in medical diagnosis.

- ○ **A.** AI medical diagnosis has no ethical implications
- ○ **B.** Medical imaging cannot be analysed by computers
- ○ **C.** AI will completely replace all doctors immediately
- ○ **D.** Benefits include potentially faster and more consistent analysis than overworked doctors; concerns include accountability if AI errors cause harm (who is responsible - doctor, hospital, or software company?), the 'black box' problem of unexplainable AI decisions, risk of over-reliance reducing doctors' diagnostic skills, and ensuring AI performance across diverse patient populations

Q26 (3 marks)

Delivery and taxi apps often classify their workers as 'independent contractors' rather than employees. Analyse the ethical and legal issues this classification raises.

- ○ **A.** Worker classification has no significant implications
- ○ **B.** Companies benefit from avoiding employment costs and obligations; workers gain flexibility but lose employment rights like minimum wage protection, sick pay, pensions, and holiday entitlement; the ethical issue is whether the actual working relationship (app control over pricing, allocation, and performance standards) constitutes employment regardless of contractual labels
- ○ **C.** All gig workers prefer being classified as contractors
- ○ **D.** Employment law does not apply to technology companies

Q27 (3 marks)

Energy companies are installing 'smart meters' that transmit detailed electricity usage data. Analyse the benefits and privacy concerns of this technology.

○ **A.** Smart meters only benefit energy companies with no concerns for consumers

○ **B.** There are no privacy concerns with smart meters

○ **C.** Smart meters cannot transmit any data

○ **D.** Benefits include accurate billing (no estimated readings), real-time usage awareness helping reduce consumption, and enabling smarter grid management; privacy concerns include detailed data revealing household activities (when people are home, sleep patterns, appliance use), potential for surveillance, data security risks, and questions about who can access usage information

Q28 (3 marks)

Some shops use facial recognition cameras to identify known shoplifters when they enter. Analyse the ethical implications of using facial recognition in retail environments.

○ **A.** Benefits include potentially reducing theft losses and improving staff safety; concerns include mass surveillance of innocent shoppers, false positive identifications causing wrongful accusations, disproportionate impact on minority groups if technology has demographic bias, normalising facial recognition surveillance, and the lack of customer consent or awareness

○ **B.** Shops can use any surveillance technology without ethical constraints

○ **C.** Facial recognition cameras cannot be used in shops

○ **D.** There are no ethical implications of retail facial recognition

Q29 (3 marks)

Companies want to test autonomous vehicles on public roads to develop the technology. Analyse the ethical issues around using public roads as

testing grounds for self-driving vehicles.

- ○ **A.** Public road testing has no ethical implications
- ○ **B.** Autonomous vehicles should never be tested anywhere
- ○ **C.** Testing is necessary for technology development but raises concerns: members of the public become unknowing participants in safety experiments; they cannot consent to the risks of sharing roads with unproven technology; early systems may cause accidents during development; and the burden of risk falls on public road users while benefits flow primarily to private companies
- ○ **D.** Autonomous vehicles are already perfectly safe and need no testing

Q30 (3 marks)

The 'right to be forgotten' allows individuals to request removal of personal information from search results. Analyse the tension between this privacy right and the public interest in information access.

- ○ **A.** The right to be forgotten has no limitations or complications
- ○ **B.** Search engines should delete everything anyone requests
- ○ **C.** All personal information should always be available forever
- ○ **D.** Individuals have legitimate interests in controlling outdated, irrelevant, or harmful information about themselves; but removal requests may conflict with public interest in accessing information about public figures, historical records, journalistic archives, and accountability for past actions - creating tension between privacy rights and freedom of information that requires case-by-case balancing

Total: **55** marks • Time: **30 min**

↑ Back to top

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